Entity Embedding

ACTL3143 & ACTL5111 Deep Learning for Actuaries

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UNSW SYDNEY

Lecture Outline

- **Entity Embedding**
- Categorical Variables & Entity Embeddings
- Keras' Functional API
- French Motor Dataset with Embeddings
- Scale By Exposure

Revisit the French motor dataset

	IDpol	ClaimNb	Exposure	Area	VehPower	VehAge	DrivAge	Bon
0	1.0	1.0	0.10000	D	5.0	0.0	55.0	50.0
1	3.0	1.0	0.77000	D	5.0	0.0	55.0	50.0
...	...	...	...	...	...	...	...	...
678011	6114329.0	0.0	0.00274	B	4.0	0.0	60.0	50.0
678012	6114330.0	0.0	0.00274	B	7.0	6.0	29.0	54.0

678013 rows × 12 columns

Source: Nell et al. (2020), Case Study: French Motor Third-Party Liability Claims, SSRN.

Data dictionary

Variable	Description	Preprocessing
`IDpol`	Policy number (unique identifier)	Dropped
`ClaimNb`	Number of claims on the given policy	**Target**
`Exposure`*	Total exposure in yearly units	Normalised
`Area`	Area code (ordinal)	Ordinal Encode
`VehPower`	Power of the car (ordinal encoded)	Normalised
`VehAge`	Age of the car in years	Normalised
`DrivAge`	Age of the (most common) driver in years	Normalised
`BonusMalus`	Bonus–malus level between 50 and 230 (with reference level 100)	Normalised
`VehBrand`*	Car brand (nominal)	One-hot
`VehGas`	Diesel or regular fuel car (binary)	One-hot
`Density`	Density of inhabitants per km^2 in the city of the living place of the driver	Normalised
`Region`*	Regions in France (prior to 2016)	One-hot

Source: Nell et al. (2020), Case Study: French Motor Third-Party Liability Claims, SSRN.

The model

Have $\{(\mathbf{x}_i, y_i)\}_{i=1,\dots,n}$ for $\mathbf{x}_i \in \mathbb{R}^{47}$ and $y_i \in \mathbb{N}_0$.

Assume the distribution

$$Y_i \sim \mathsf{Poisson}(\lambda(\mathbf{x}_i))$$

We have $\mathbb{E}Y_i = \lambda(\mathbf{x}_i)$. The NN takes $\mathbf{x}_i$ & predicts $\mathbb{E}Y_i$.

> **Note**
>
> For insurance, *this is a bit weird*. The exposures are different for each policy.
>
> $\lambda(\mathbf{x}_i)$ is the expected number of claims for the duration of policy i's contract.
>
> Normally, $\text{Exposure}_i \not\in \mathbf{x}_i$, and $\lambda(\mathbf{x}_i)$ is the expected rate *per year*, then
>
> $$Y_i \sim \mathsf{Poisson}(\text{Exposure}_i \times \lambda(\mathbf{x}_i)).$$

What values do we see in the data?

```python
X_train_raw["Area"].value_counts()
X_train_raw["VehBrand"].value_counts()
X_train_raw["VehGas"].value_counts()
X_train_raw["Region"].value_counts()
```

```
Area
C    5514
D    4116
     ...
B    2387
F     444
Name: count, Length: 6, dtype: int64
```

```
VehBrand
B1     4998
B2     4906
       ...
B11     283
B14     140
Name: count, Length: 11, dtype: int64
```

```
VehGas
Regular    10658
Diesel      8092
Name: count, dtype: int64
```

```
Region
R24    6493
R82    2112
       ...
R42      48
R43      26
Name: count, Length: 22, dtype: int64
```

How we preprocessed last time

```python
from sklearn.compose import make_column_transformer

ct = make_column_transformer(
  (OneHotEncoder(sparse_output=False, drop="first"), ["VehGas", "VehBrand", "Region"]),
  (OrdinalEncoder(), ["Area"]),
  remainder=StandardScaler(),
  verbose_feature_names_out=False
)
X_train = ct.fit_transform(X_train_raw)
```

```python
X_train_raw.head(3)
```

	Exposure	Area	VehPower	VehAge
0	1.00	A	7.0	8.0
1	0.79	B	7.0	7.0
2	1.00	C	6.0	13.0

```python
X_train.head(3)
```

	VehGas_Regular	VehBrand_B10	Ve
0	0.0	0.0	0.(
1	0.0	0.0	0.(
2	1.0	0.0	0.(

3 rows × 39 columns

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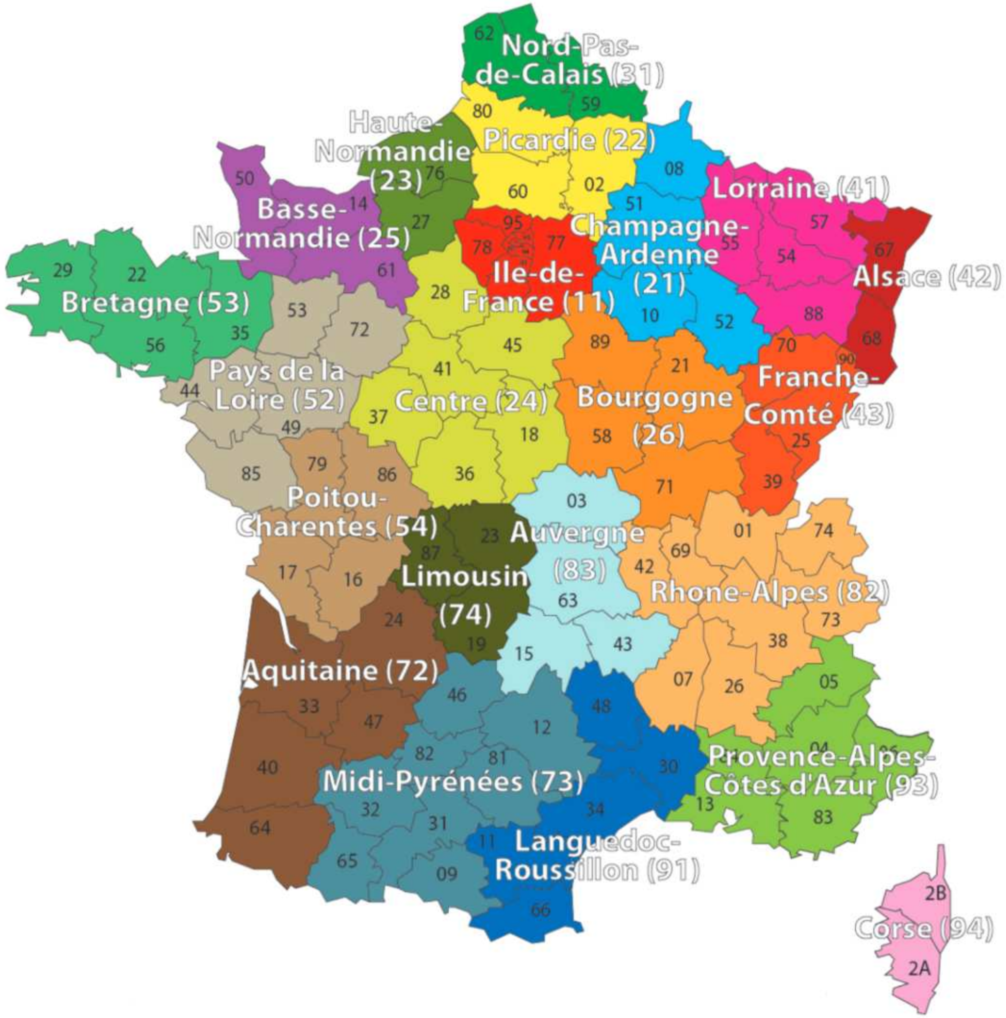
Region column



French Administrative Regions

Source: Wikimedia

One-hot encoding

```python
oh = OneHotEncoder(sparse_output=False)
X_train_oh = oh.fit_transform(X_train_raw[["Region"]])
X_test_oh = oh.transform(X_test_raw[["Region"]])
print(list(X_train_raw["Region"][:5]))
X_train_oh.head()
```

```
['R24', 'R21', 'R53', 'R24', 'R82']
```

	Region_R11	Region_R21	Region_R22	Region_R23	Region_R24	Region_R2
0	0.0	0.0	0.0	0.0	1.0	0.0
1	0.0	1.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...
3	0.0	0.0	0.0	0.0	1.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0

5 rows × 22 columns

Train on one-hot inputs

```python
num_regions = len(oh.categories_[0])

random.seed(12)
model = Sequential([
  Dense(2, input_dim=num_regions),
  Dense(1, activation="exponential")
])

model.compile(optimizer="adam", loss="poisson")

es = EarlyStopping(verbose=True)
hist = model.fit(X_train_oh, y_train, epochs=100, verbose=0,
    validation_split=0.2, callbacks=[es])
hist.history["val_loss"][-1]
```

```
Epoch 6: early stopping
```

```
0.7679625153541565
```

Make a fake batch of data

```
X = np.eye(num_regions)
pd.DataFrame(X, columns=oh.categories_[0]
```

	R11	R21	R22	R23	R24	R25	R2
0	1.0	0.0	0.0	0.0	0.0	0.0	0.(
1	0.0	1.0	0.0	0.0	0.0	0.0	0.(
…	…	…	…	…	…	…	…
20	0.0	0.0	0.0	0.0	0.0	0.0	0.(
21	0.0	0.0	0.0	0.0	0.0	0.0	0.(

22 rows × 22 columns

```
model.layers[0](X)
```

```
tensor([[-0.0646, -0.2346],
        [ 0.1463, -0.0323],
        [-0.3050, -0.2913],
        [ 0.0257, -0.3189],
        [ 0.3848, -0.2778],
        [ 0.0983, -0.4137],
        [ 0.2235, -0.2502],
        [ 0.1192,  0.3202],
        [ 0.6295, -0.4385],
        [ 0.2008, -0.0306],
        [-0.2030, -0.3797],
        [ 0.7612, -0.2256],
        [ 0.5068, -0.0436],
        [ 0.7867, -0.4326],
        [-0.2026, -0.1788],
        [-0.2430, -0.0941],
```

The first layer

```python
layer = model.layers[0]
W, b = layer.get_weights()
X.shape, W.shape, b.shape
```

```
((22, 22), (22, 2), (2,))
```

```python
X @ W + b
```

```
array([[-0.06, -0.23],
       [ 0.15, -0.03],
       [-0.3 , -0.29],
       [ 0.03, -0.32],
       [ 0.38, -0.28],
       [ 0.1 , -0.41],
       [ 0.22, -0.25],
       [ 0.12,  0.32],
       [ 0.63, -0.44],
       [ 0.2 , -0.03],
       [-0.2 , -0.38],
       [ 0.76, -0.23],
       [ 0.51, -0.04],
       [ 0.79, -0.43],
       [-0.2 , -0.18],
       [-0.24, -0.09],
```

```python
W + b
```

```
array([[-0.06, -0.23],
       [ 0.15, -0.03],
       [-0.3 , -0.29],
       [ 0.03, -0.32],
       [ 0.38, -0.28],
       [ 0.1 , -0.41],
       [ 0.22, -0.25],
       [ 0.12,  0.32],
       [ 0.63, -0.44],
       [ 0.2 , -0.03],
       [-0.2 , -0.38],
       [ 0.76, -0.23],
       [ 0.51, -0.04],
       [ 0.79, -0.43],
       [-0.2 , -0.18],
       [-0.24, -0.09],
```

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Just a look-up operation

```
display(list(oh.categories_[0]))
```

```
['R11',
 'R21',
 'R22',
 'R23',
 'R24',
 'R25',
 'R26',
 'R31',
 'R41',
 'R42',
 'R43',
 'R52',
 'R53',
 'R54',
 'R72',
 'R73',
```

```
W + b
```

```
array([[-0.06, -0.23],
       [ 0.15, -0.03],
       [-0.3 , -0.29],
       [ 0.03, -0.32],
       [ 0.38, -0.28],
       [ 0.1 , -0.41],
       [ 0.22, -0.25],
       [ 0.12,  0.32],
       [ 0.63, -0.44],
       [ 0.2 , -0.03],
       [-0.2 , -0.38],
       [ 0.76, -0.23],
       [ 0.51, -0.04],
       [ 0.79, -0.43],
       [-0.2 , -0.18],
       [-0.24, -0.09],
```

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Turn the region into an index

```
oe = OrdinalEncoder()
X_train_reg = oe.fit_transform(X_train_raw[["Region"]])
X_test_reg = oe.transform(X_test_raw[["Region"]])

for i, reg in enumerate(oe.categories_[0][:3]):
  print(f"The Region value {reg} gets turned into {i}.")
```

```
The Region value R11 gets turned into 0.
The Region value R21 gets turned into 1.
The Region value R22 gets turned into 2.
```

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Use an Embedding layer

```
from keras.layers import Embedding
num_regions = X_train_raw["Region"].nunique()

random.seed(12)
model = Sequential([
  Embedding(input_dim=num_regions, output_dim=2),
  Dense(1, activation="exponential")
])

model.compile(optimizer="adam", loss="poisson")
```

```
es = EarlyStopping(verbose=True)
hist = model.fit(X_train_reg, y_train, epochs=100, verbose=0,
    validation_split=0.2, callbacks=[es])
hist.history["val_loss"][-1]
```

```
Epoch 4: early stopping
```

```
0.7677991390228271
```

```
model.layers
```

```
[<Embedding name=embedding, built=True>, <Dense name=dense_2, built=True>]
```

Keras' Embedding Layer

```
model.layers[0].get_weights()[0]
```

```
array([[ 0.05, -0.07],
       [-0.02,  0.03],
       [-0.04, -0.02],
       [ 0.09, -0.12],
       [ 0.24, -0.22],
       [ 0.16, -0.19],
       [ 0.17, -0.17],
       [-0.05,  0.09],
       [ 0.36, -0.33],
       [ 0.03, -0.01],
       [ 0.02, -0.07],
       [ 0.36, -0.3 ],
       [ 0.21, -0.16],
       [ 0.42, -0.37],
       [-0.02, -0.02],
       [-0.06,  0.02],
```

```
X_train_raw["Region"].head(4)
```

```
0    R24
1    R21
2    R53
3    R24
Name: Region, dtype: str
```

```
X_sample = X_train_reg[:4].to_numpy()
X_sample
```

```
array([[ 4.],
       [ 1.],
       [12.],
       [ 4.]])
```

```
enc_tensor = model.layers[0](X_sample)
keras.ops.convert_to_numpy(enc_tensor).sc
```

```
array([[ 0.24, -0.22],
       [-0.02,  0.03],
       [ 0.21, -0.16],
       [ 0.24, -0.22]], dtype=float32)
```

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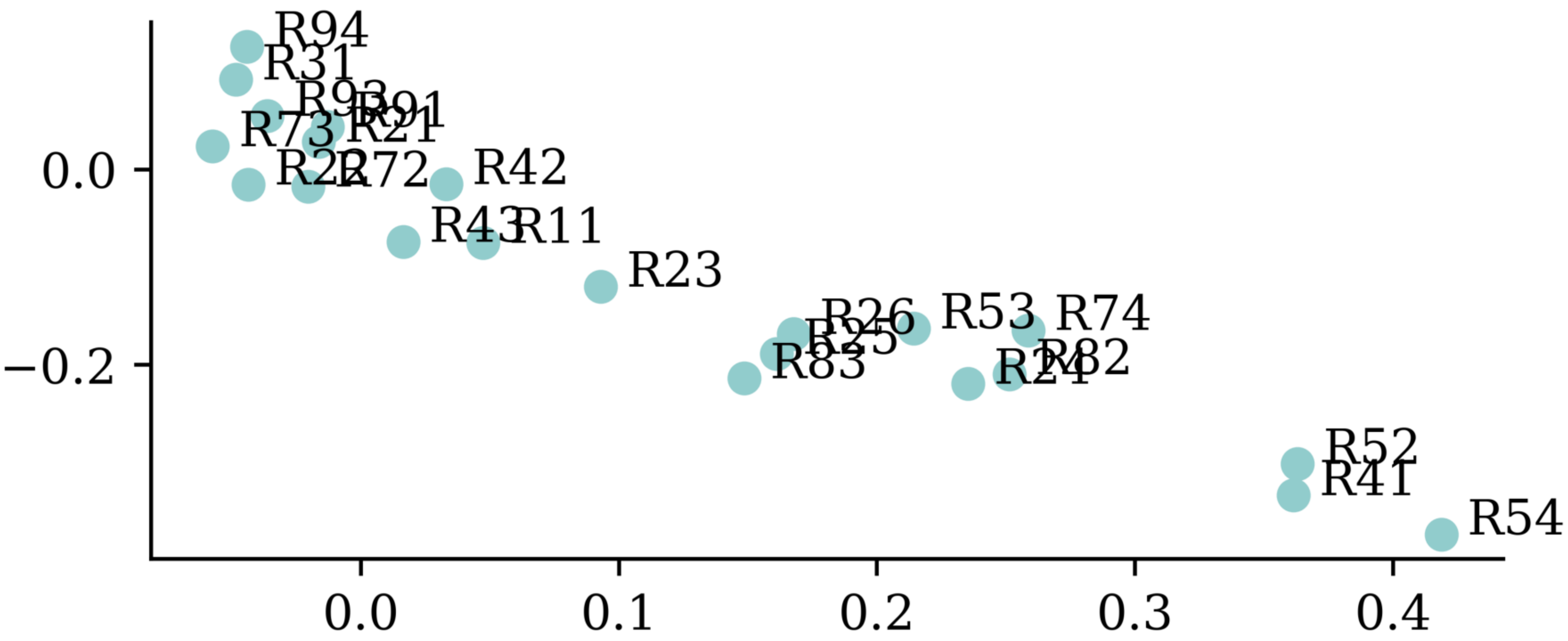
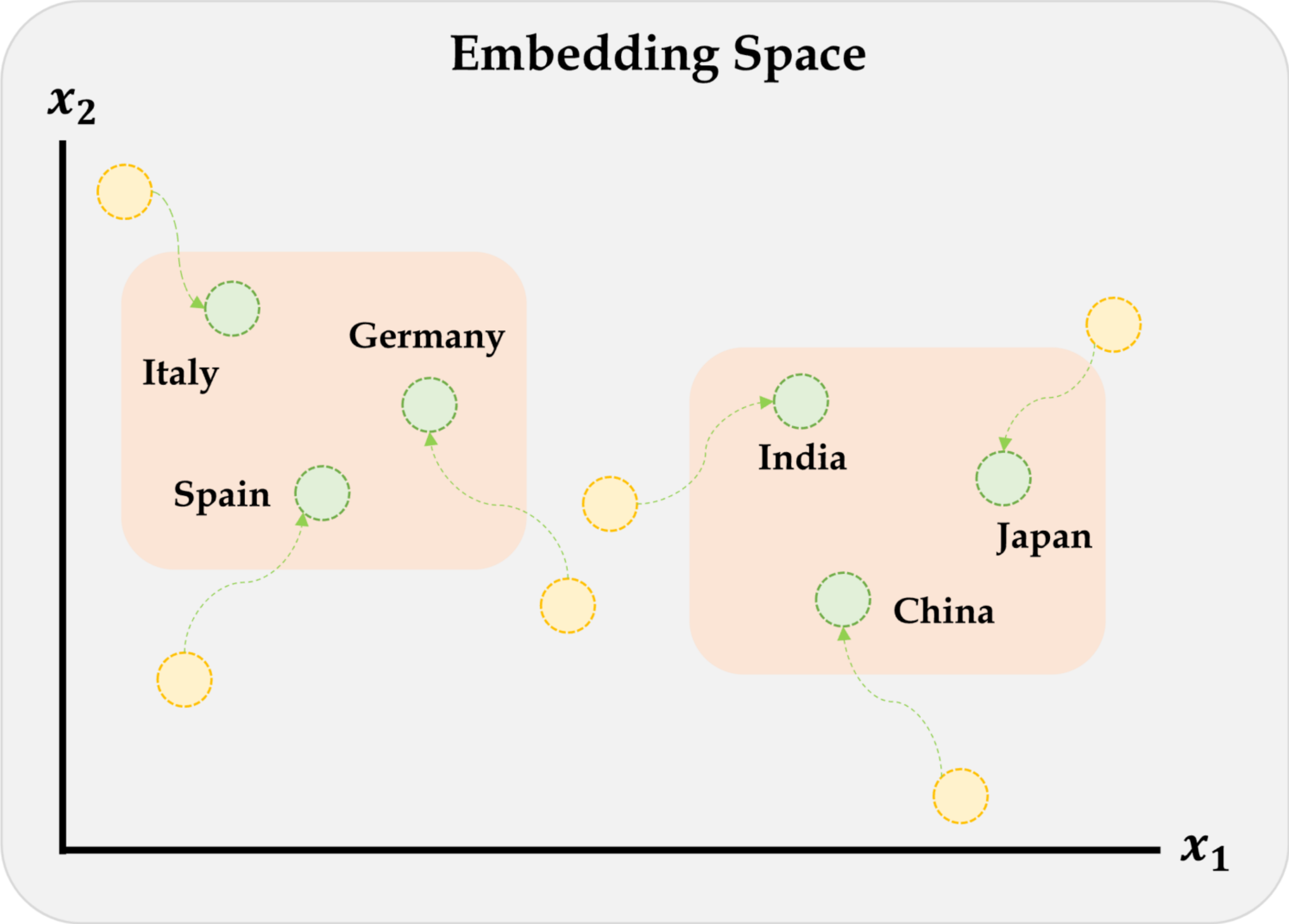
The learned embeddings

```
points = model.layers[0].get_weights()[0]
plt.scatter(points[:,0], points[:,1])
for i in range(num_regions):
  plt.text(points[i,0]+0.01, points[i,1] , s=oe.categories_[0][i])
```



Entity embeddings



Embeddings will gradually improve during training.

Source: Marcus Lautier (2022).

Embeddings & other inputs

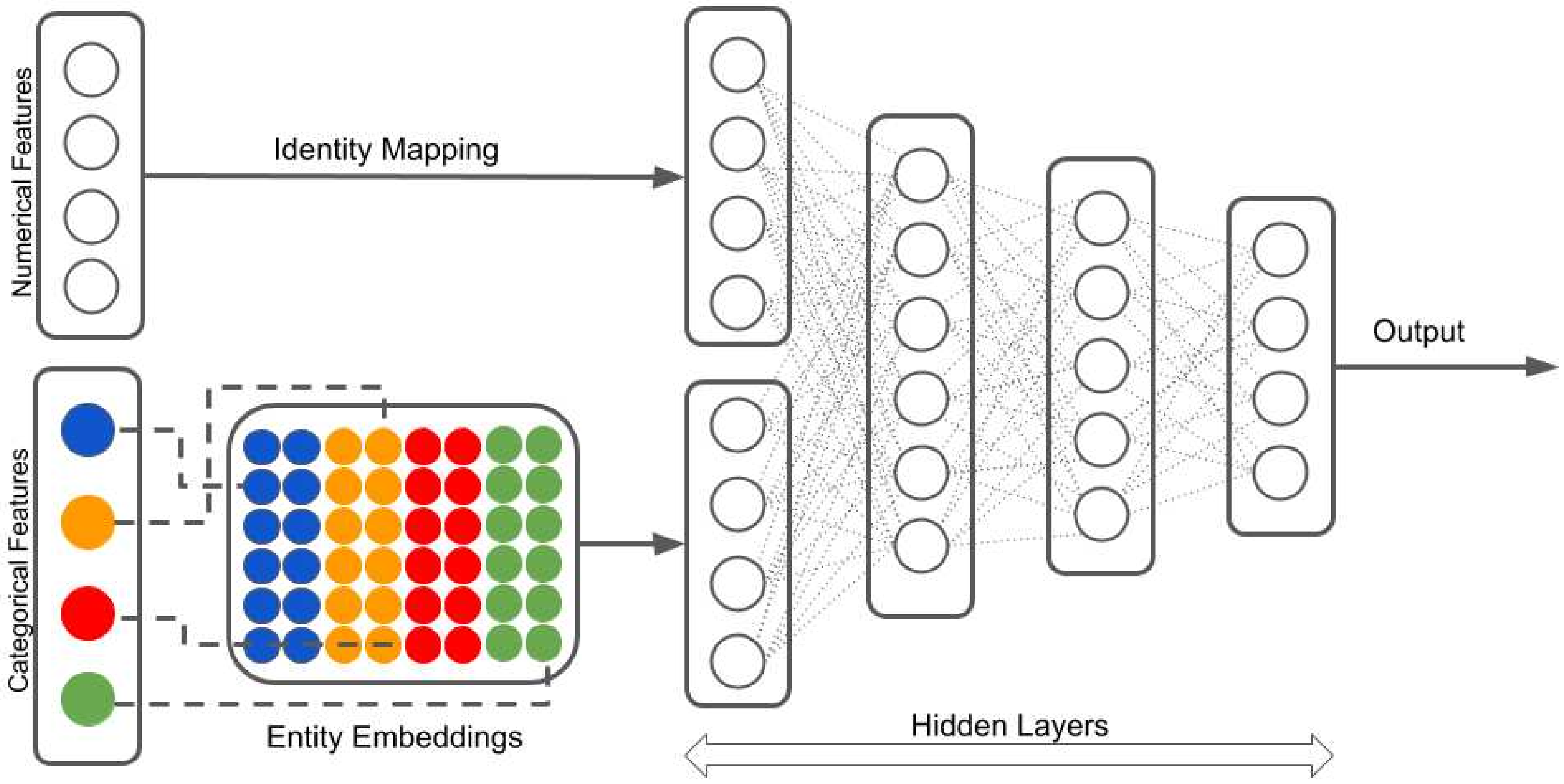


Illustration of a neural network with both continuous and categorical inputs.

We can't do this with Sequential models...

Source: LotusLabs Blog, Accurate insurance claims prediction with Deep Learning.

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Converting Sequential models

```python
from keras.models import Model
from keras.layers import Input
```

```python
random.seed(12)

model = Sequential([
  Dense(30, "leaky_relu"),
  Dense(1, "exponential")
])

model.compile(
  optimizer="adam",
  loss="poisson")

hist = model.fit(
  X_train_oh, y_train,
  epochs=1, verbose=0,
  validation_split=0.2)
hist.history["val_loss"][-1]
```

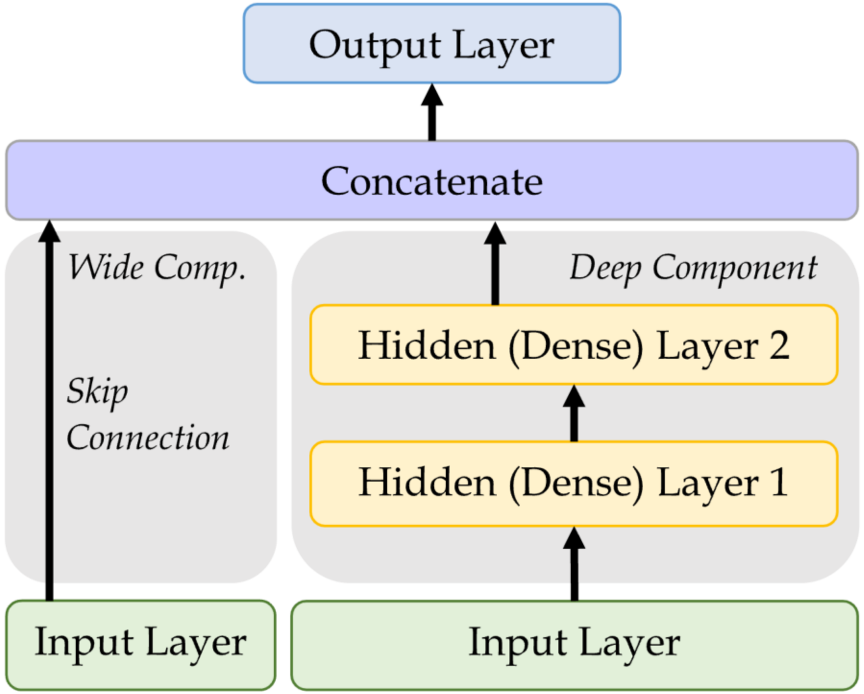
```
0.7695862650871277
```

```python
random.seed(12)

inputs = Input(shape=(X_train_oh.shape[1]
x = Dense(30, "leaky_relu")(inputs)
out = Dense(1, "exponential")(x)
model = Model(inputs, out)

model.compile(
  optimizer="adam",
  loss="poisson")

hist = model.fit(
  X_train_oh, y_train,
  epochs=1, verbose=0,
  validation_split=0.2)
hist.history["val_loss"][-1]
```

```
0.769521236419677
```

See one-length tuples.

Wide & Deep network



An illustration of the wide & deep network architecture.

Add a *skip connection* from input to output layers.

```python
from keras.layers \
    import Concatenate

inp = Input(shape=X_train.shape[1:])
hidden1 = Dense(30, "leaky_relu")(inp)
hidden2 = Dense(30, "leaky_relu")(hidden1)
concat = Concatenate()(
  [inp, hidden2])
output = Dense(1)(concat)
model = Model(
    inputs=[inp],
    outputs=[output])
```

Naming the layers

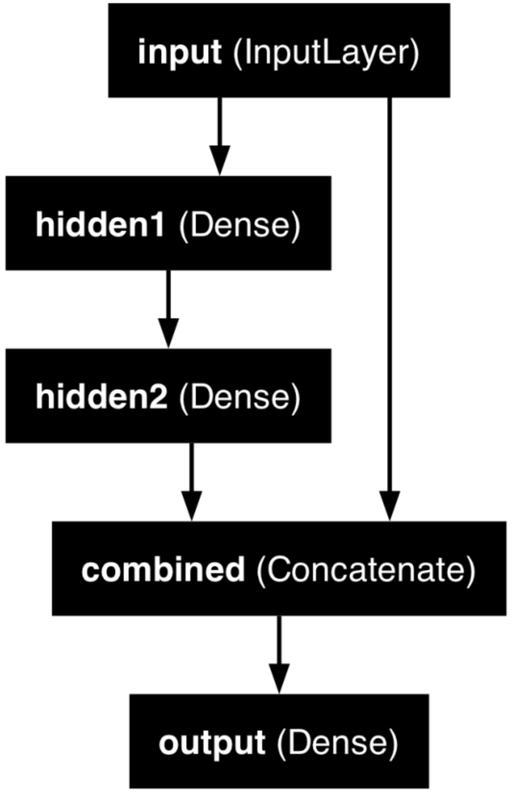
For complex networks, it is often useful to give meaningful names to the layers.

```
input_ = Input(shape=X_train.shape[1:], name="input")
hidden1 = Dense(30, activation="leaky_relu", name="hidden1")(input_)
hidden2 = Dense(30, activation="leaky_relu", name="hidden2")(hidden1)
concat = Concatenate(name="combined")([input_, hidden2])
output = Dense(1, name="output")(concat)
model = Model(inputs=[input_], outputs=[output])
```

Inspecting a complex model

```python
from keras.utils import plot_model
```

```python
plot_model(model, show
```




```python
model.summary(line_length=75)
```

Model: "functional_5"

Layer (type)	Output Shape	Param #	Connected
input (InputLayer)	(None, 39)	0	-
hidden1 (Dense)	(None, 30)	1,200	input[0][0
hidden2 (Dense)	(None, 30)	930	hidden1[0]
combined (Concatenate)	(None, 69)	0	input[0][0 hidden2[0]
output (Dense)	(None, 1)	70	combined[0

Total params: 2,200 (8.59 KB)
Trainable params: 2,200 (8.59 KB)
Non-trainable params: 0 (0.00 B)

Lecture Outline

The desired architecture

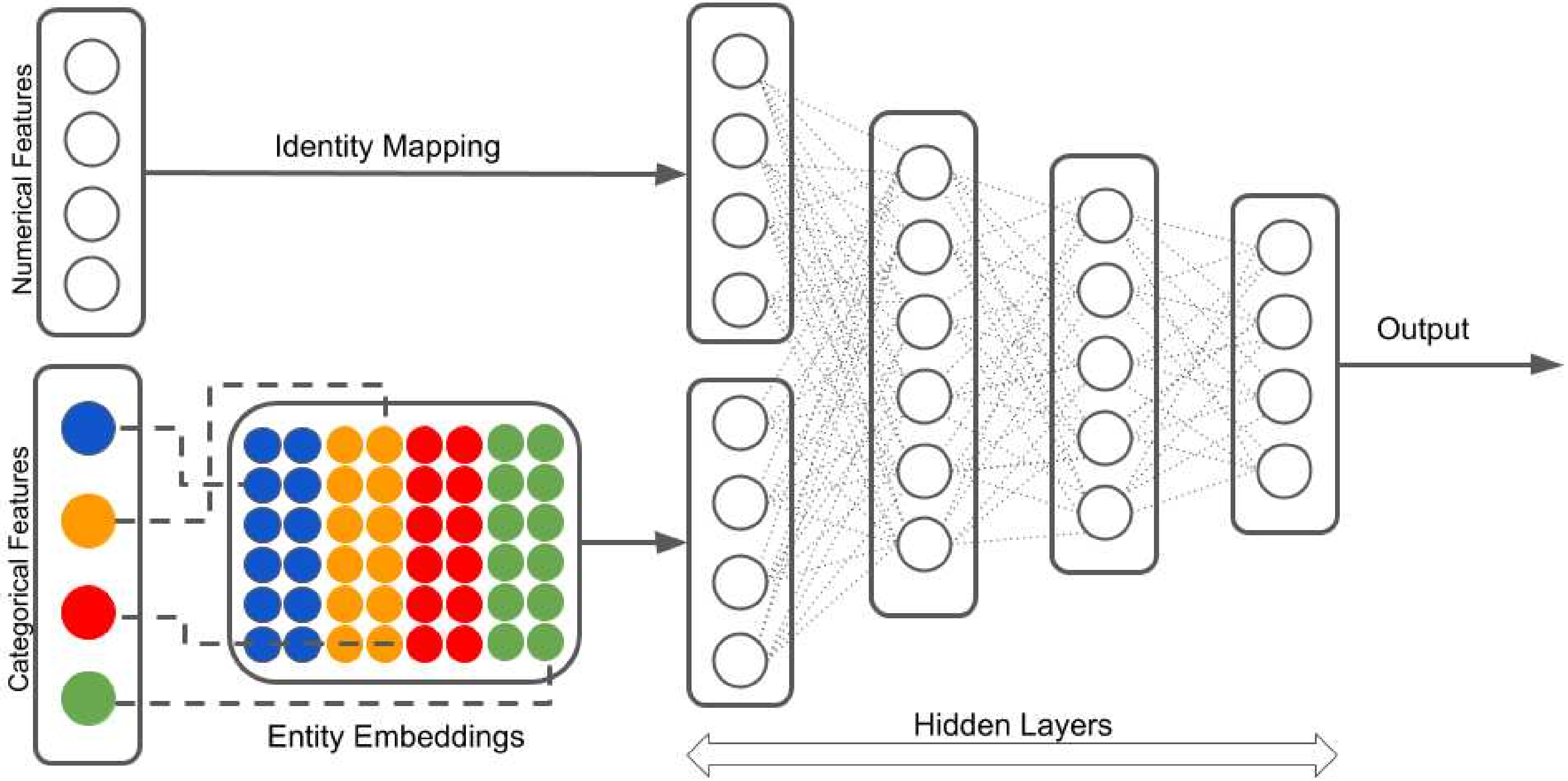


Illustration of a neural network with both continuous and categorical inputs.

Source: LotusLabs Blog, Accurate insurance claims prediction with Deep Learning.

Preprocess all French motor inputs

Transform the categorical variables to integers:

```
num_brands, num_regions = X_train_raw[["VehBrand", "Region"]].nunique()

ct = make_column_transformer(
  (OrdinalEncoder(), ["VehBrand", "Region", "Area", "VehGas"]),
  remainder=StandardScaler(),
  verbose_feature_names_out=False
)
X_train = ct.fit_transform(X_train_raw)
X_test = ct.transform(X_test_raw)
```

Split the brand and region data apart from the rest:

```
X_train_brand = X_train["VehBrand"]
X_train_region = X_train["Region"]
X_train_rest = X_train.drop(["VehBrand", "Region"], axis=1)

X_test_brand = X_test["VehBrand"]
X_test_region = X_test["Region"]
X_test_rest = X_test.drop(["VehBrand", "Region"], axis=1)
```

Organise the inputs

Make a Keras `Input` for: vehicle brand, region, & others.

```
veh_brand = Input(shape=(1,), name="veh_brand")
region = Input(shape=(1,), name="region")
other_inputs = Input(shape=X_train_rest.shape[1:], name="other_inputs")
```

Create embeddings and join them with the other inputs.

```
from keras.layers import Reshape

random.seed(1337)
veh_brand_ee = Embedding(input_dim=num_brands, output_dim=2,
    name="veh_brand_ee")(veh_brand)
veh_brand_ee = Reshape(target_shape=(2,))(veh_brand_ee)

region_ee = Embedding(input_dim=num_regions, output_dim=2,
    name="region_ee")(region)
region_ee = Reshape(target_shape=(2,))(region_ee)

x = Concatenate(name="combined")([veh_brand_ee, region_ee, other_inputs])
```

Complete the model and fit it

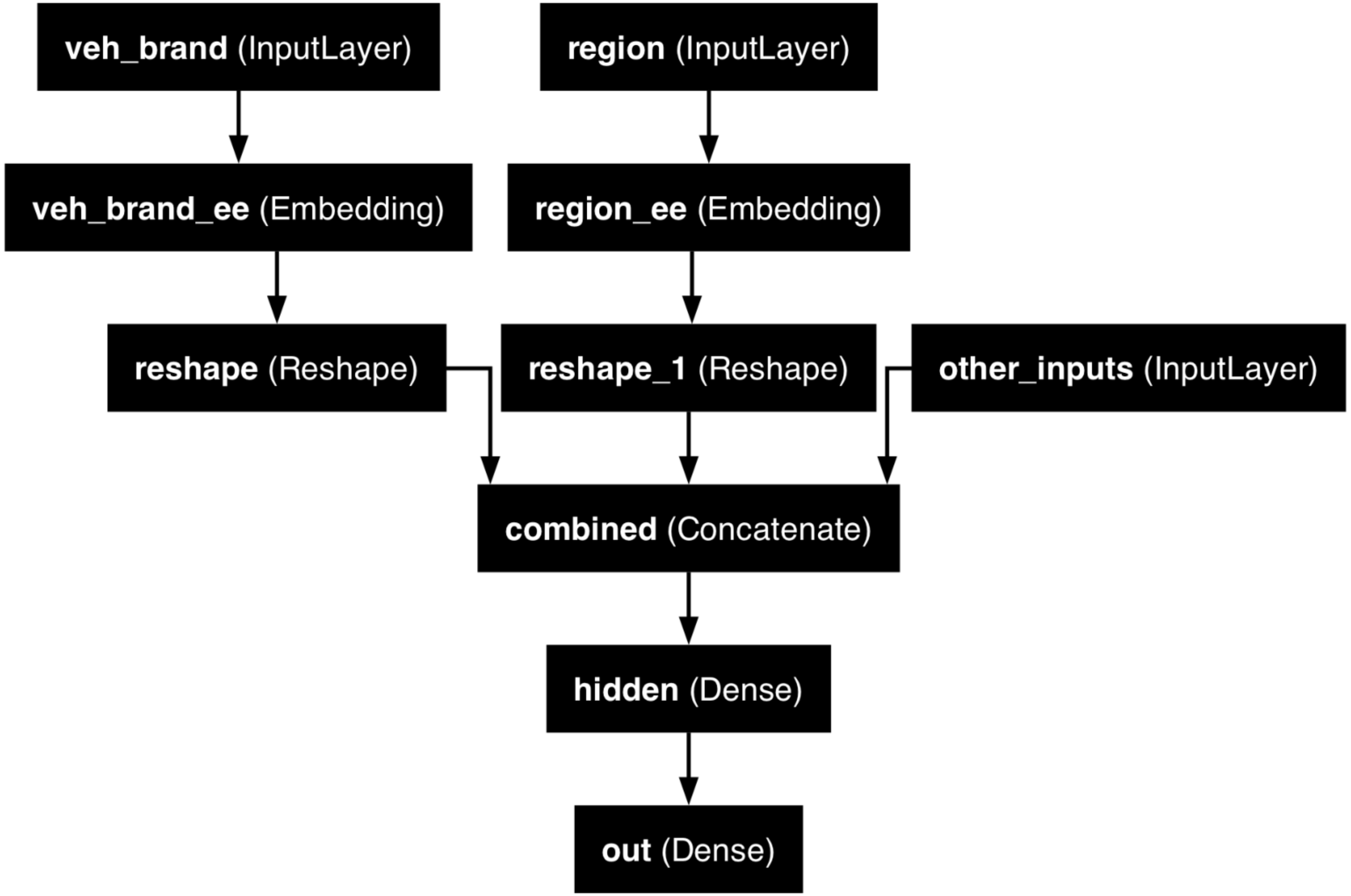
Feed the combined embeddings & continuous inputs to some normal dense layers.

```
x = Dense(30, "relu", name="hidden")(x)
out = Dense(1, "exponential", name="out")(x)

model = Model([veh_brand, region, other_inputs], out)
model.compile(optimizer="adam", loss="poisson")

hist = model.fit((X_train_brand, X_train_region, X_train_rest),
    y_train, epochs=100, verbose=0,
    callbacks=[EarlyStopping(patience=5)], validation_split=0.2)
np.min(hist.history["val_loss"])
```

```
np.float64(0.6855398416519165)
```

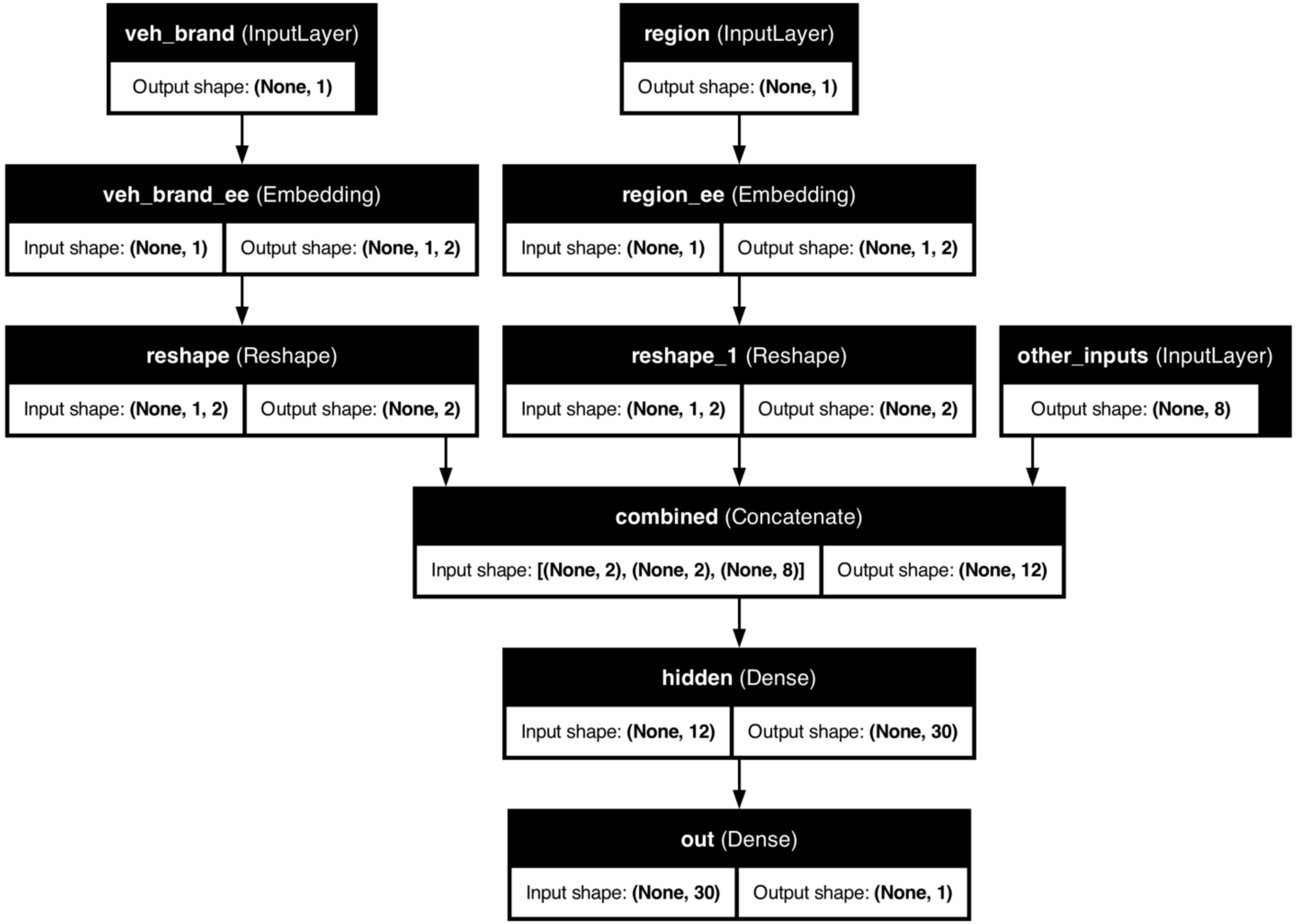
Plotting this model

```python
plot_model(model, show_layer_names=True)
```



Why we need to reshape

```python
plot_model(model, show_layer_names=True, show_shapes=True)
```



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Two different models

Have $\{(\mathbf{x}_i, y_i)\}_{i=1,\dots,n}$ for $\mathbf{x}_i \in \mathbb{R}^{47}$ and $y_i \in \mathbb{N}_0$.

Model 1: Say $Y_i \sim \mathsf{Poisson}(\lambda(\mathbf{x}_i))$.

But, the exposures are different for each policy. $\lambda(\mathbf{x}_i)$ is the expected number of claims for the duration of policy i's contract.

Model 2: Say $Y_i \sim \mathsf{Poisson}(\text{Exposure}_i \times \lambda(\mathbf{x}_i))$.

Now, $\text{Exposure}_i \not\in \mathbf{x}_i$, and $\lambda(\mathbf{x}_i)$ is the rate *per year*.

Just take continuous variables

```
ct = make_column_transformer(
  ("passthrough", ["Exposure"]),
  ("drop", ["VehBrand", "Region", "Area", "VehGas"]),
  remainder=StandardScaler(),
  verbose_feature_names_out=False
)
X_train = ct.fit_transform(X_train_raw)
X_test = ct.transform(X_test_raw)
```

Split exposure apart from the rest:

```
X_train_exp = X_train["Exposure"]
X_test_exp = X_test["Exposure"]
X_train_rest = X_train.drop("Exposure", axis=1)
X_test_rest = X_test.drop("Exposure", axis=1)
```

Organise the inputs:

```
exposure = Input(shape=(1,), name="exposure")
other_inputs = Input(shape=X_train_rest.shape[1:], name="other_inputs")
```

Make & fit the model

Feed the continuous inputs to some normal dense layers.

```python
random.seed(1337)
x = Dense(30, "relu", name="hidden1")(other_inputs)
x = Dense(30, "relu", name="hidden2")(x)
lambda_ = Dense(1, "exponential", name="lambda")(x)
```

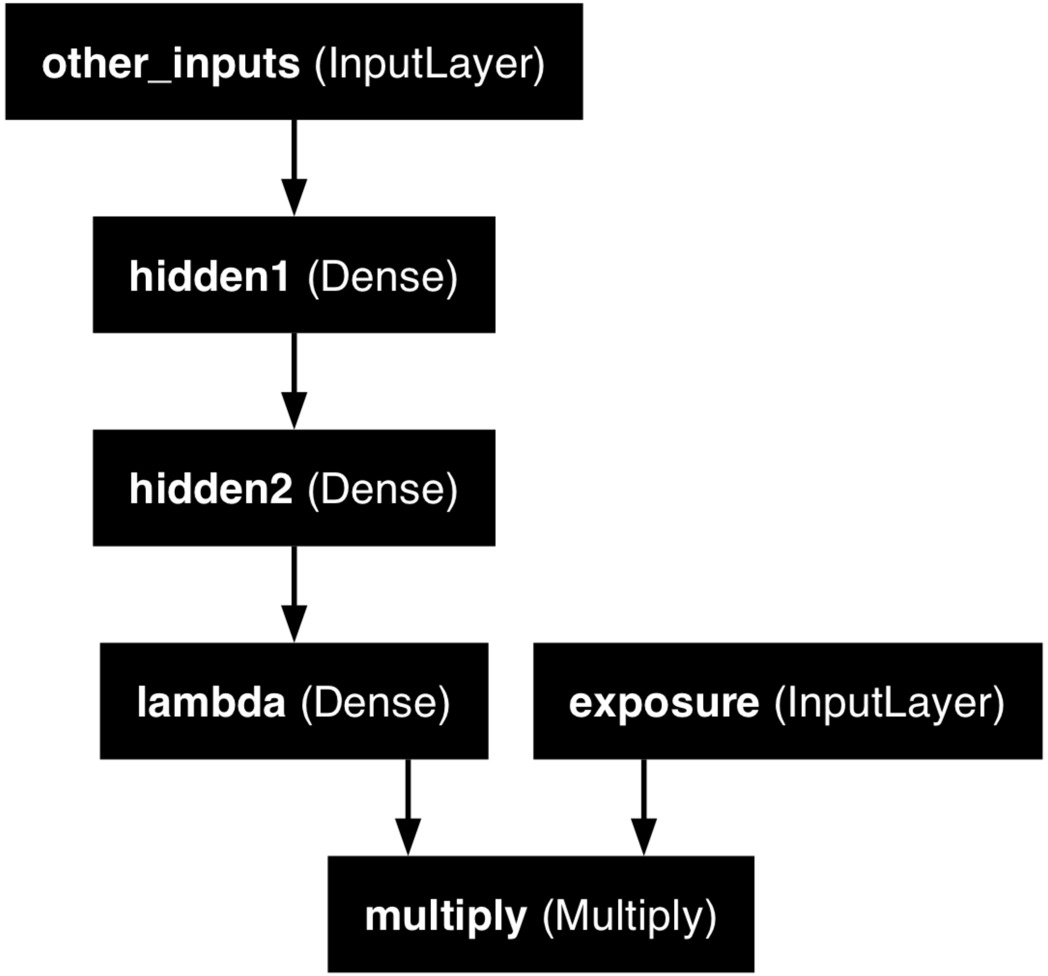
```python
out = lambda_ * exposure # In past, need keras.layers.Multiply()[lambda_, exposure]
model = Model([exposure, other_inputs], out)
model.compile(optimizer="adam", loss="poisson")

es = EarlyStopping(patience=10, restore_best_weights=True, verbose=1)
hist = model.fit((X_train_exp, X_train_rest),
    y_train, epochs=100, verbose=0,
    callbacks=[es], validation_split=0.2)
np.min(hist.history["val_loss"])
```

```
Epoch 29: early stopping
Restoring model weights from the end of the best epoch: 19.
```

```
np.float64(0.9100556373596191)
```

Plot the model

```
plot_model(model, show_layer_names=True)
```



Package Versions

```python
from watermark import watermark
print(watermark(python=True, packages="keras,matplotlib,numpy,pandas,seaborn,scipy,torch"
```

```
Python implementation: CPython
Python version       : 3.14.5
IPython version      : 9.13.0

keras     : 3.14.1
matplotlib: 3.10.9
numpy     : 2.4.4
pandas    : 3.0.2
seaborn   : 0.13.2
scipy     : 1.17.1
torch     : 2.11.0
```

Glossary

- entity embeddings
- Input layer
- Keras functional API
- Reshape layer
- skip connection
- wide & deep network